5C34 : ADVANCED AI
EMOTION DRIVEN MUSIC GENERATOR

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Slide 1: Where We Left Off

At our last presentation, we had a plan:

- Build a multi-modal AI that converts how you're feeling into music
- Use pre-trained models from Hugging Face rather than training from scratch
- Two-model architecture: **DistilBERT** (emotion) $\rightarrow$ **MusicGen** (audio)

Today, we have something better: we have a working demo.

We built it, refined it, and gave it a name: **Catharsis**.

A Greek origin word meaning emotional release through art

Slide 2: Catharsis is Live

What exists right now, in our Jupyter Notebook:

- Both models running on Google Colab T4 GPU
- Full end-to-end pipeline: type a sentence → receive a `.wav` file
- Gradio interface deployed with a live public URL
- Emotion confidence breakdown shown to the user
- User controls for duration (5–30 seconds) and creativity (temperature)

Key Point

Catharsis is no longer just a concept. It is a working, deployed application with an end-to-end user flow.

Slide 3: The Pipeline We Built

The complete data flow inside the notebook:

```
User types a sentence
↓
DistilBERT Emotion Classifier
(returns confidence scores for all 6 emotions)
↓
Emotion Analyzer
(single emotion OR multi-emotion blend?)
↓
Prompt Construction Engine
(selects or blends musical descriptors)
↓
MusicGen (facebook/musicgen-small)
(autoregressive audio token generation)
↓
EnCodec Decoder → 32kHz .wav file
↓
Gradio Interface (playable audio + emotion breakdown)
```

Slide 4: Stage 1 – Emotion Classification

Model: `bhadresh-savani/distilbert-base-uncased-emotion` (66M parameters)

What it does:

- Takes the user's raw text as input
- Passes it through DistilBERT's self-attention layers
- Outputs a softmax probability distribution across **6 emotion labels**: Joy, Sadness, Anger, Fear, Love, Surprise

What we get back:

Example

Input: `"I passed my exam but I already miss studying for it"`
Output: {joy: 0.71, sadness: 0.26, surprise: 0.02, . . . }

Key design choice: We retrieve **all six scores**, not just the top one.

Slide 5: Stage 2 – The Emotion Analyzer

The insight that shaped our entire architecture:

Human emotions are rarely singular. The same sentence can carry two feelings simultaneously. Our `analyzeEmotions` function detects this:

- If one emotion dominates (secondary scores $< 20\%$), it is classified as single emotion mode
- If a secondary emotion exceeds 20% confidence, it is classified as mixed emotion mode

Input	**Distribution**	**Mode**
"I'm so happy today!"	Joy: 0.94	Single $\rightarrow$ Joy
"I passed but I miss the journey"	Joy: 0.71, Sadness: 0.29	Mixed $\rightarrow$ Joy + Sadness
"Excited but terrified about tomorrow"	Joy: 0.65, Fear: 0.31	Mixed $\rightarrow$ Joy + Fear

Slide 6: Stage 3 — Prompt Construction: Single Emotion

For **single emotions**, we built a confidence-aware prompt selector.

The same emotion at different intensities should sound different. So rather than one flat label mapping to one string, each emotion has **three prompt variants** tied to confidence levels:

Confidence	**Anger Prompt Generated**
> 0.8 (High)	*"Intense rock with heavy electric guitar and powerful aggressive drums"*
0.5–0.8 (Medium)	*"Dramatic orchestral with brass and strong percussion"*
< 0.5 (Low)	*"Firm piano with purposeful, determined rhythm"*

The music scales with how strongly the user feels that emotion.

Slide 7: Stage 3 — Prompt Construction: Mixed Emotions

For **mixed emotions**, we built `EMOTION_BLENDS`, which is a curated dictionary of emotional pairs mapped to crafted musical descriptions:

Emotion Pair	**Generated Music Prompt**
Joy + Sadness	*"Bittersweet piano ballad mixing hope and melancholy"*
Joy + Fear	*"Anxious yet excited electronic, tense builds and euphoric releases"*
Love + Sadness	*"Emotional ballad expressing longing and devotion simultaneously"*
Anger + Love	*"Passionate rock ballad, love and frustration intertwined"*
Surprise + Fear	*"Suspenseful music with unexpected twists, thriller undertones"*

Each prompt was **manually tested and refined** to capture the nuance of the emotional blend.

Slide 8: Stage 4 — Audio Generation (MusicGen)

Model: `facebook/musicgen-small` (300M parameters, T4 GPU)

What happens after the prompt is constructed:

1. The prompt string is tokenized and encoded by **T5's encoder** (MusicGen's text backbone)
2. T5 embeddings are passed as conditioning into the **autoregressive decoder** via cross-attention
3. The decoder predicts discrete **audio tokens** sequentially, token by token
4. The **EnCodec decoder** converts discrete tokens → continuous **32kHz stereo waveform**

Parameters we expose to the user:

- `max_new_tokens = duration × 50` → controls length
- `temperature` (0.7 to 1.3) → controls creativity
- `guidance_scale = 3.0` → how strictly to follow the prompt

Slide 9: The Catharsis Interface

What the user sees:

- A text box: *"Enter your text — a journal entry, a message, how you're feeling..."*
- **Duration** slider: 5 to 30 seconds
- **Creativity** slider: conservative (0.7) to experimental (1.3)
- Output: playable audio widget + downloadable `.wav`

Emotion breakdown panel (shown after generation):

Example Output

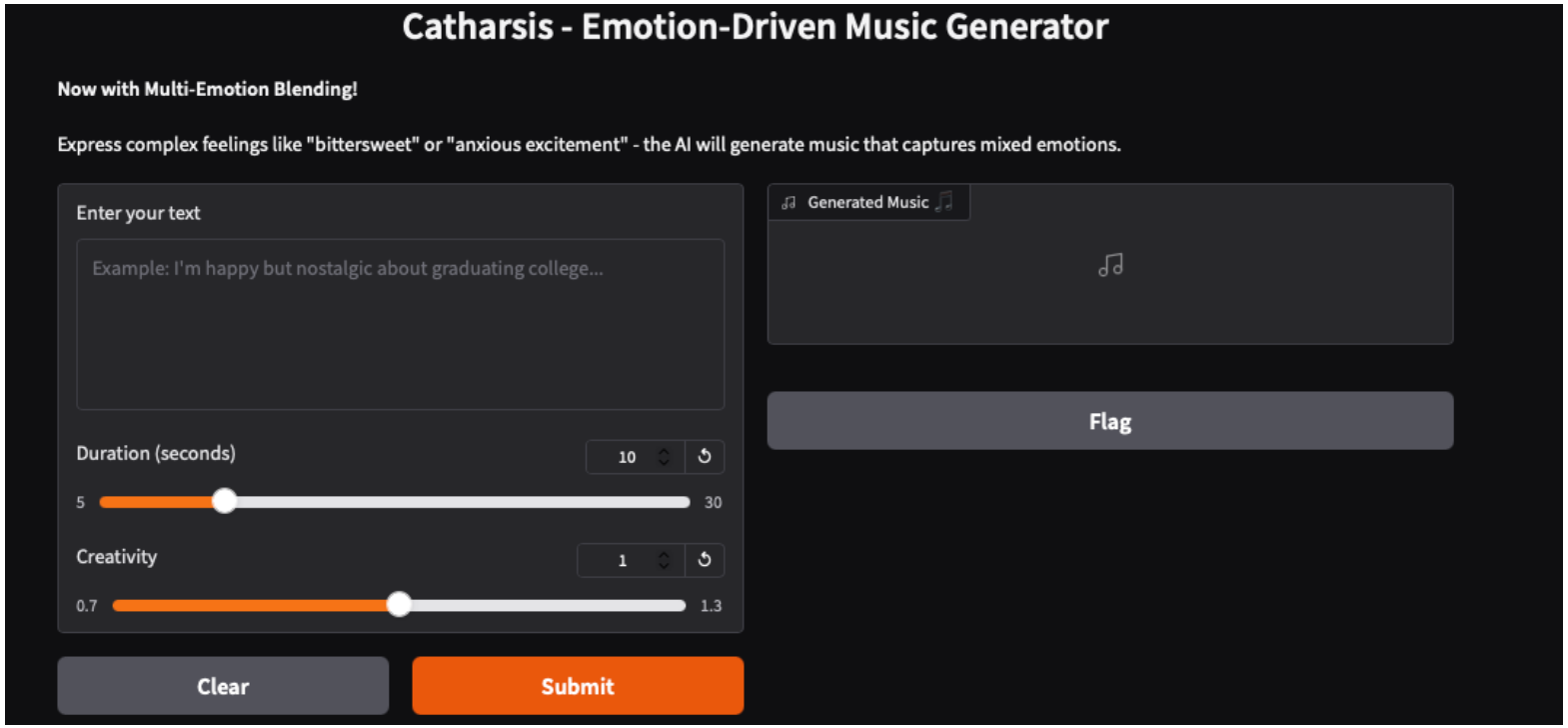
Detected Emotions:	joy / sadness
Confidence:	joy: 0.71 | sadness: 0.29
Music Style:	bittersweet piano ballad mixing hope & melancholy

Pre-loaded example prompts:

- *"I finally achieved my dream, but I miss the journey getting here."*
- *"I'm excited but terrified about my first day at the new job tomorrow!"*
- *"Walking alone at night, every shadow makes my heart race with dread."*

Slide 10: Live Demo

This is the interface of Catharsis.



Slide 11: Challenges & What Comes Next

Challenges we hit:

- *musicgen-medium* exceeds the T4 GPU's VRAM, hence we are bound to *musicgen-small*
- Generation latency: $\sim$15–20 seconds per 10-second clip (autoregressive bottleneck)
- MusicGen is highly sensitive to exact prompt phrasing and required extensive manual tuning
- DistilBERT cannot detect sarcasm or irony (*"Oh great, another Monday"* $\rightarrow$ flags as Joy)

What we're building next:

- Replace *EMOTION_BLENDS* with an **LLM Prompt Expander** (*TinyLlama-1.1B*) to dynamically generate unique music prompts per input
- Conduct a **user study**: ask participants if generated music matches the emotion of their text
- Evaluate audio quality using **Fréchet Audio Distance (FAD)**
- Deploy Catharsis permanently on **Hugging Face Spaces**

Thank you!

Because your feelings deserve a soundtrack!

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